Mohamed Adeep

DevOps Engineer | Cloud & Infrastructure Automation | Kubernetes | CI/CD | Terraform

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Professional Summary

DevOps Engineer with **2.5+ years** of experience at Tata Consultancy Services (TCS) in cloud infrastructure management, CI/CD automation, containerization, infrastructure as code, and application deployment. Experienced with **AWS, Azure, GCP, Linux, Terraform, Ansible, Docker, Kubernetes, Jenkins, GitLab CI/CD, and GitOps** workflows. Skilled in automating infrastructure provisioning, managing containerized applications, implementing monitoring and alerting, troubleshooting deployment issues, and improving infrastructure reliability. Reduced environment setup time by **70%** and saved approximately **8 hours** of manual effort per week through automation. Strong understanding of cloud networking, security, deployment automation, observability, and DevOps best practices.

Technical Skills

Cloud Platforms: AWS, Microsoft Azure, Google Cloud Platform (GCP), Hybrid Cloud, On-Premises Infrastructure
AWS Services: EC2, EKS, ECS, ECR, S3, VPC, IAM, RDS, Lambda, ALB, NLB, CloudWatch, Route 53, AWS Certificate Manager
Azure Services: Azure Kubernetes Service (AKS), Virtual Machines, Virtual Networks (VNets), Azure Monitor
GCP Services: Google Kubernetes Engine (GKE), Compute Engine, Cloud Storage, Artifact Registry
Infrastructure as Code: Terraform, Reusable Terraform Modules, Remote State Management, Ansible, AWS CloudFormation, AWS CDK, Azure Bicep
Containers and Orchestration: Docker, Kubernetes, Amazon EKS, Azure AKS, Google GKE, OpenShift, Helm, Deployments, Services, Ingress, ConfigMaps, Secrets, RBAC
CI/CD and GitOps: Jenkins, Jenkins Pipelines, GitLab CI/CD, GitHub Actions, Azure DevOps, Harness, Argo CD, Argo Workflows
Version Control: Git, GitHub, GitLab, Branching Strategies, Code Integration
Operating Systems: Linux, Ubuntu, RHEL, Amazon Linux, Windows Server, IIS Administration, Server Patching
Scripting and Automation: Bash, Shell Scripting, Python, PowerShell
Monitoring and Logging: Prometheus, Grafana, Alertmanager, ELK Stack, AWS CloudWatch, Datadog, Dynatrace, Splunk
Networking: TCP/IP, DNS, VPC, VNets, Routing, Load Balancing, SSL/TLS, VPN, Firewalls, Security Groups
DevSecOps and Security: SonarQube, Trivy, Snyk, Gitleaks, IAM, Secrets Management, Vulnerability Scanning, Security Quality Gates
Reliability and Operations: Autoscaling, High Availability, Incident Management, Root Cause Analysis, Backup and Recovery, Capacity Monitoring, Cloud Cost Optimization
Tools and Practices: Jira, ServiceNow, Agile/Scrum, GitOps, Change Management, Technical Documentation, Operational Runbooks

Professional Experience

DevOps Engineer **Mar 2025 – Present**
Tata Consultancy Services (TCS) – Project: Healthcare Application Cloud & DevOps Platform

- Managed AWS infrastructure, including **EC2, VPC, IAM, S3, EKS, and CloudWatch**, to support application availability and reliability.
- Built and maintained **15+ CI/CD pipelines** using Jenkins, Git, and GitHub to automate application builds, testing, and deployments.
- Containerized application microservices using **Docker** and deployed **15+ workloads** on Kubernetes/EKS.
- Automated infrastructure provisioning with **Terraform** and server configuration with **Ansible**, reducing environment setup time by **70%**.
- Supported hybrid and multi-cloud infrastructure across AWS, Azure, GCP, and on-premises environments.
- Developed **GitLab CI/CD** workflows to execute Terraform infrastructure changes and Ansible configuration tasks.
- Implemented GitOps-based deployment workflows using **Argo CD and Helm** for version-controlled application deployments.
- Configured AWS Application Load Balancers, Network Load Balancers, and Kubernetes Ingress to support application traffic routing.
- Managed **SSL/TLS certificate** issuance, renewal, and rotation using AWS Certificate Manager and configured TLS termination.
- Administered Kubernetes resources, including Deployments, Services, Ingress, ConfigMaps, Secrets, Namespaces, and RBAC.
- Automated deployments, health checks, backups, and routine operational tasks using **Bash and Linux scripting**,

saving approximately **8 hours** of manual effort per week.

- Implemented monitoring and alerting using **Prometheus, Grafana, and CloudWatch**, improving issue detection by **40%**.
- Configured centralized logging using **ELK Stack** to support troubleshooting and root cause analysis.
- Integrated security scanning and code quality checks into CI/CD pipelines, contributing to a reported **35%** reduction in vulnerabilities.
- Supported **10+ monthly releases** with a reported **95% release success rate**.
- Maintained technical documentation, operational runbooks, and deployment procedures to support incident management and reliable operations.

Junior DevOps Engineer **Mar 2024 – Feb 2025**

Tata Consultancy Services (TCS) – Project: Digital Banking Cloud & DevOps

- Managed AWS infrastructure resources, including **EC2, VPC, IAM, S3, EKS, and CloudWatch**, across application environments.
- Developed and maintained **10+ CI/CD pipelines** using Jenkins, Git, and GitHub to automate application builds, testing, and releases.
- Containerized microservices using **Docker** and supported Kubernetes/EKS deployments using **Helm**.
- Provisioned cloud infrastructure using **Terraform**, reducing environment setup time by **60%**.
- Automated server configuration and application deployment across **15+ Linux servers** using Ansible and shell scripting, reducing manual effort by **50%**.
- Supported **Windows Server** administration, including service management, configuration checks, and patching using PowerShell.
- Implemented monitoring and centralized logging using **Prometheus, Grafana, CloudWatch, and ELK Stack**.
- Managed Kubernetes resources, including Deployments, Services, Ingress, ConfigMaps, Secrets, and RBAC.
- Integrated security scanning and quality checks into CI/CD workflows to improve deployment consistency and security.
- Automated rolling deployments, health checks, and environment provisioning to improve release reliability.
- Assisted with deployment troubleshooting, configuration management, and operational automation.

Projects

End-to-End DevSecOps CI/CD Pipeline on AWS EKS | *AWS, Jenkins, Git, Docker, Kubernetes, Terraform, SonarQube,*

- Developed a Jenkins pipeline to automate source code checkout, testing, security scanning, container image creation, and application deployment.
- Integrated code quality analysis and vulnerability scanning into the CI/CD workflow.
- Implemented rolling updates, automated rollback procedures, and post-deployment smoke tests.
- Improved consistency and repeatability of the application delivery process.

Kubernetes Monitoring and Observability Stack | *Python, Kubernetes, Prometheus, Grafana, Helm, kind*

- Instrumented a Python application with custom metrics to monitor application health and service performance.
- Configured **10 alert rules** and a **15-panel** Grafana dashboard for monitoring and troubleshooting.
- Implemented CI validation of the monitoring stack on a local Kubernetes cluster using kind.
- Improved visibility into application behavior through metrics, dashboards, and alerting.

Multi-Cloud Infrastructure as Code | *Terraform, Ansible, AWS, Azure, Linux*

- Developed **six reusable Terraform modules** to provision repeatable development, staging, and production environments.
- Configured remote Terraform state management to support infrastructure provisioning workflows.
- Created Ansible roles for Linux configuration, patching, firewall rules, and SSH hardening.
- Standardized infrastructure configuration to reduce repetitive manual provisioning tasks.

Key Achievements

- Reduced infrastructure environment setup time by **70%** through Terraform automation.
- Saved approximately **8 hours** of manual effort per week through operational automation.
- Supported more than **10 releases per month** with a reported **95% release success rate**.
- Improved issue detection by **40%** through monitoring and alerting improvements.
- Contributed to a reported **35%** reduction in vulnerabilities through automated security and quality gates.

Certifications and Training

AWS Certified Cloud Practitioner – Amazon Web Services **Sep 2025**

AWS re/Start Program – Cloud Computing, Networking, Security, and Linux **Jul 2025 – Aug 2025**

Education

HKRH College, Madurai Kamaraj University **2022 – 2025**

Bachelor of Science in Computer Science – CGPA: 7.2/10 *Theni, Tamil Nadu*

Additional Information

Notice Period: Immediate Joiner | **Relocation:** Open to Relocation | **Languages:** English, Tamil